

Thread - 2025-11-08

<https://x.com/RiikonenMarko/status/1987113829096910903>

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On the Finnish niche halo records. Timo Nevala just set a new record for the deepest-winter 18° plate arcs, and in the photographically confirmed category also for 23° plate arcs on the night of 6/7 Nov 2025 in Kurikka. <https://t.co/UWvGAICYky> This is 45 nights from the winter solstice night of 21/22 Dec. The earlier record holder Martikainen's observation on 6/7 February 2023 in Juva was beaten by two nights as it is 47 nights from the winter solstice. <https://t.co/qlPvKuayM>

In Finland we see pyramid stuff in high clouds in winter, but they are just 9° halos. So these deep-winter 18 and 23° arcs are of interest.

The most deepest deep-winter 23° plate arc observation in Finnish archives is a lunar display on the night of 30/31 Dec 1998 in Savukoski by Simo Aikioniemi, discussed on page 9 in Ursa Minor 1/1999 <https://t.co/qR3RIXb8HH> (the details of the drawing have greatly suffered in reproduction). The display had also 9 halo and 24° plate arc – the latter lasting half a minute and marked uncertain by the observer. I am penchant on relegating this in the "uncertain" category as it is not nearly always easy to distinguish 23° plate arc from 22° tanarc visually, lunar displays being even more fraught with misidentification.

The Finnish odd radius plate arc displays, at least those containing 23° and 18°, have been regarded to appear more likely during warmer than average weather periods. This notion has concerned the summer-half of the year, but if we look at these three winter-half observations, the warmer-than-average holds for them, too.

On the night Martikainen made his observation, the temp visited at -10 C at the lowest, but next day hiked up above 0° C and this state of affairs lasted until the 10th. The January-February average for the nearby Mikkeli is -7 C.

And Nevala's observation was at the tail end of an exceptional heat wave during which all-time November records were broken at several stations.

There are no data available for Savukoski for 1998. But at least 130 km south-west in Rovaniemi a heat-wave hit on the day following the observation, lifting the temps in the 0°...-5 C range for more than a week.

As Martikainen bags odd radii in his stacks from pretty much any odd display, I wonder how the weather statistics analysis would out. Maybe the connection to warmer weather periods would crumble. But I would say it still has meaning for the visual displays.
